

INITIAL TECHNICAL REPORT

Carnot Limit in Mechanical Systems and Practical Efficiency Boundaries

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Executive Summary

The Carnot limit defines the maximum theoretical efficiency possible for a heat-based mechanical system operating between two temperature boundaries. In practical engineering, this limit should not be treated as an achievable operating target. Instead, it should be treated as a reference boundary that helps engineers evaluate whether a proposed design is physically realistic.

Most real systems operate significantly below Carnot efficiency because practical machines contain unavoidable losses such as friction, turbulence, heat leakage, pressure drop, electrical conversion loss and fouling. Therefore, the real engineering challenge is not reaching Carnot efficiency, but understanding why the gap exists and how it can be reduced safely and economically.

This report explains the Carnot limit using engineering-oriented interpretation, practical examples, testing methodology, thermal analysis, risk assessment and design guidance relevant to pumps, compressors, thermal systems and purification-related devices.

1. Introduction

1.1 Objective of This Report

The objective of this report is to explain the Carnot limit from the perspective of practical engineering design and efficiency analysis. The discussion focuses on how the Carnot concept should guide realistic efficiency expectations, system benchmarking and design optimization.

1.2 Why Carnot Efficiency Matters

- It defines the absolute theoretical upper efficiency boundary governed by the second law of thermodynamics, making it the irreducible reference for any heat-based system.
- It prevents unrealistic efficiency claims during design reviews and proposal evaluations, protecting the project from technically invalid performance targets.
- It helps identify dominant irreversibility sources in real systems by quantifying the gap between theoretical and measured performance, directing corrective engineering effort precisely.
- It supports rigorous efficiency benchmarking during acceptance testing and commissioning, providing an objective thermodynamic standard independent of competitor claims.
- It guides thermal system optimisation by revealing which temperature boundary modifications deliver the highest marginal efficiency improvement per unit of engineering investment.
- It enables cross-system comparison across different machine types and scales, allowing engineers to rank design alternatives on a common thermodynamic basis regardless of working fluid or cycle configuration.
- It strengthens technical risk assessments by flagging designs where claimed efficiency approaches or exceeds the Carnot limit, triggering mandatory review of system boundary definitions and sensor calibration integrity.

1.3 Scope and Boundaries

This report focuses on heat-based mechanical systems including heat engines, compressors, thermal control systems, pumps and purification-related systems where heat transfer or thermal exchange affects efficiency.

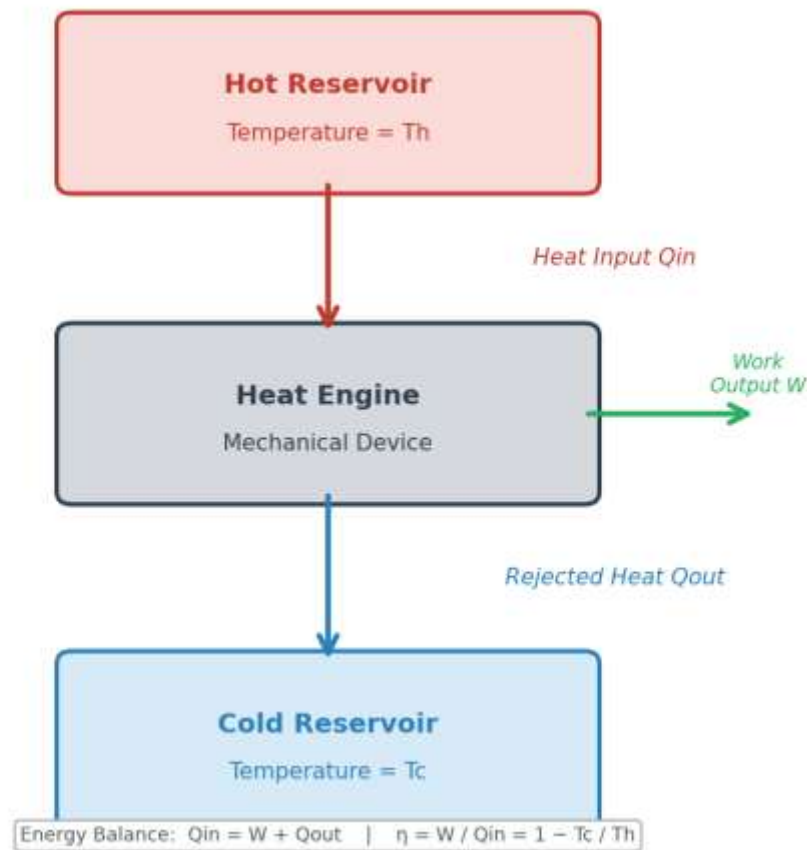


Figure 1. Basic heat-engine energy flow representation.

2. System Requirements and Constraints

2.1 Functional Requirements

- Measure hot-side temperature accurately at the thermal boundary using calibrated sensors traceable to a national metrology standard, with uncertainty quantified and recorded in the test log.
- Measure cold-side or environmental sink temperature at the actual heat rejection interface, not at a remote ambient location, to prevent systematic underestimation of T_c and consequent overestimation of Carnot efficiency.
- Measure useful output energy or mechanical work at the defined system output boundary, excluding parasitic loads such as cooling fans, control electronics and lubrication pumps unless they form part of the declared system scope.
- Measure total input energy at the system inlet, accounting for all energy streams including electrical, thermal, pressure-enthalpy and chemical, to ensure the efficiency denominator is physically complete.

- Calculate Carnot efficiency using absolute Kelvin temperatures and compute actual efficiency from measured input-output data, then derive the irreversibility ratio as the quotient of actual to Carnot efficiency for normalised comparison.
- Compare actual efficiency against the Carnot benchmark and express the efficiency gap as both an absolute percentage and a relative fraction, enabling prioritisation of loss-reduction efforts by engineering impact.
- Generate a structured loss attribution report identifying the fractional contribution of each identified loss mechanism friction, heat leakage, pressure drop, electrical conversion and fouling to the total efficiency gap, providing the design team with a ranked improvement roadmap.

2.2 Non-Functional Requirements

Constraint	Meaning for the System
Reliability	Measurements should remain repeatable
Safety	High temperatures and electrical systems must remain controlled
Environmental Stability	Ambient conditions should be recorded
Maintainability	Sensors and moving parts should be easy to inspect
Scalability	Efficiency logic should work for larger systems
Data Quality	Incorrect sensor readings create misleading results

2.3 Important Engineering Constraint

The Carnot equation must always use absolute temperature in Kelvin. Using Celsius values directly produces physically incorrect efficiency values.

3. Technical Background and Prior Art

3.1 What Is the Carnot Limit?

The Carnot limit is the maximum theoretical efficiency of a heat engine operating between a hot reservoir and a cold reservoir. It comes from the second law of thermodynamics, which states that some energy must always be rejected to a colder sink.

3.2 Carnot Efficiency Formula

$$\eta = 1 - T_c / T_h$$

Where T_c is the cold-side temperature in Kelvin and T_h is the hot-side temperature in Kelvin.

3.3 Practical Engineering Interpretation

The equation shows that efficiency increases when the temperature gap increases. However, increasing hot-side temperature also increases thermal stress, corrosion rate, sealing difficulty and safety risk.

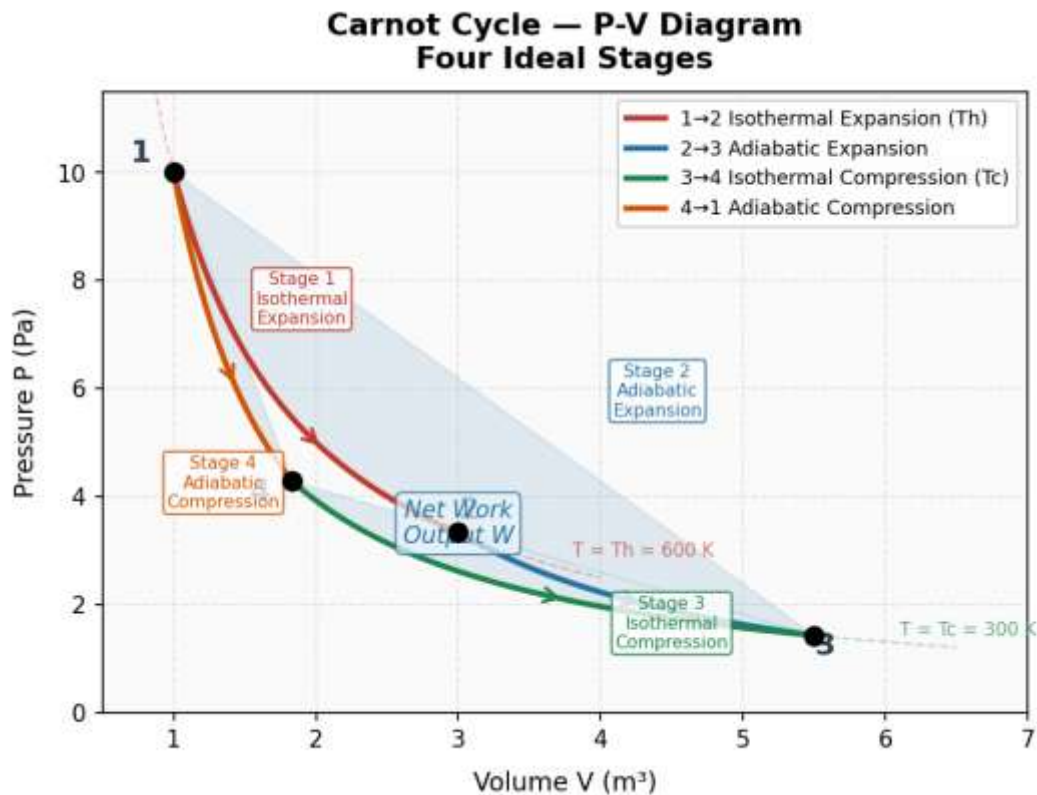


Figure 2. Four ideal stages of a Carnot cycle.

4. Detailed Technical Architecture

4.1 Conceptual Architecture for Efficiency Analysis

Layer	Main Components	Purpose
Physical Layer	Pump, motor, compressor, heat source	Performs actual work
Measurement Layer	Temperature, pressure and power sensors	Collects data
Analysis Layer	Carnot calculations and loss analysis	Converts measurements into insight
Decision Layer	Maintenance and control optimization	Improves long-term performance

4.2 Interfaces and Data Flow

The practical workflow should measure temperature boundaries, calculate Carnot efficiency, measure actual efficiency and identify dominant losses.



Figure 3. Workflow for applying Carnot analysis during design and testing.

4.3 Practical Architecture Interpretation

The Carnot limit acts like a warning boundary. If measured efficiency exceeds the Carnot limit, either the calculation is wrong or the system boundary has been defined incorrectly.

5. Design Calculations and Engineering Analysis

5.1 Example Calculation

Assume a system receives heat at 600 K and rejects heat to a 300 K environment.

$$\eta = 1 - 300/600 = 0.50 = 50\%$$

This means 50% is the theoretical upper limit. Real systems normally achieve much lower values because of unavoidable losses.

5.2 Temperature Sensitivity

The Carnot equation is highly sensitive to temperature difference. Even small increases in hot-side temperature can significantly improve theoretical efficiency.

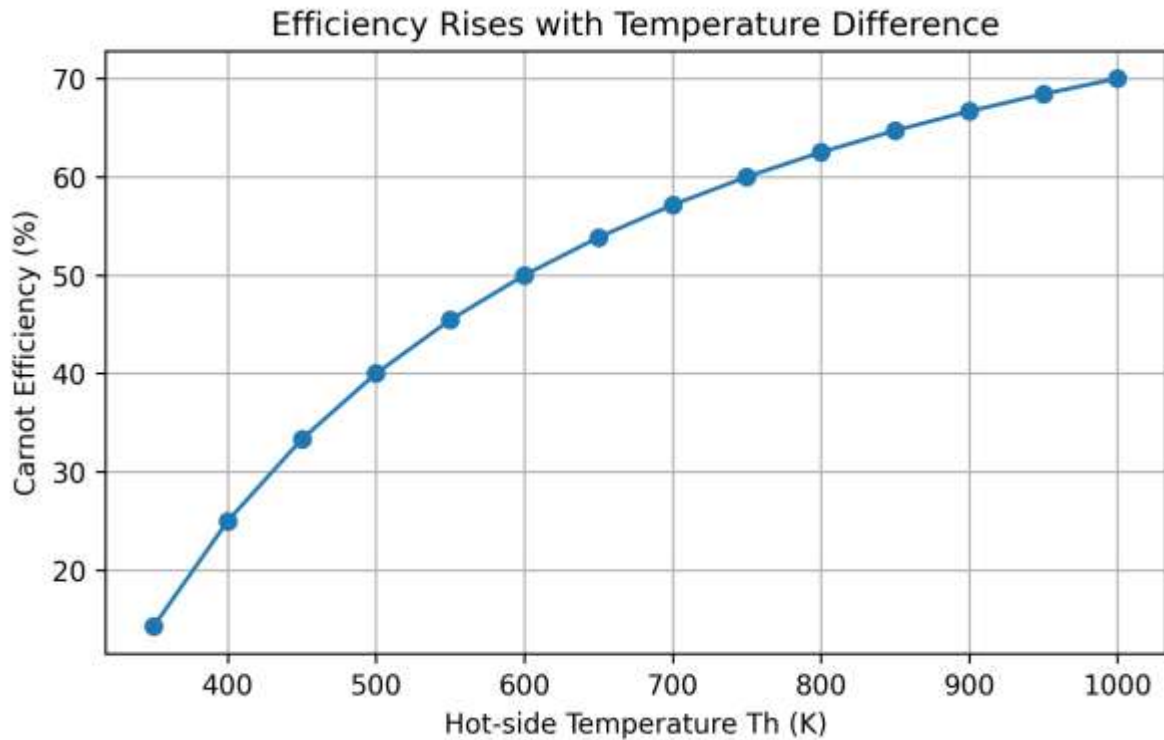


Figure 4. Carnot efficiency curve when cold-side temperature remains fixed.

6. Why Real Systems Operate Below Carnot Efficiency

Real systems are irreversible because they contain friction, turbulence, heat leakage, electrical losses and fouling.

Loss Source	What Happens	Effect on Efficiency
Friction	Moving parts rub together	Useful work decreases
Heat Leakage	Heat escapes through walls	Less heat available for conversion
Pressure Drop	Flow faces resistance	More pumping energy required
Electrical Loss	Motors/controllers are imperfect	Input energy increases
Fouling	Deposits build up	Heat transfer degrades
Sensor Error	Incorrect measurements	Control quality reduces

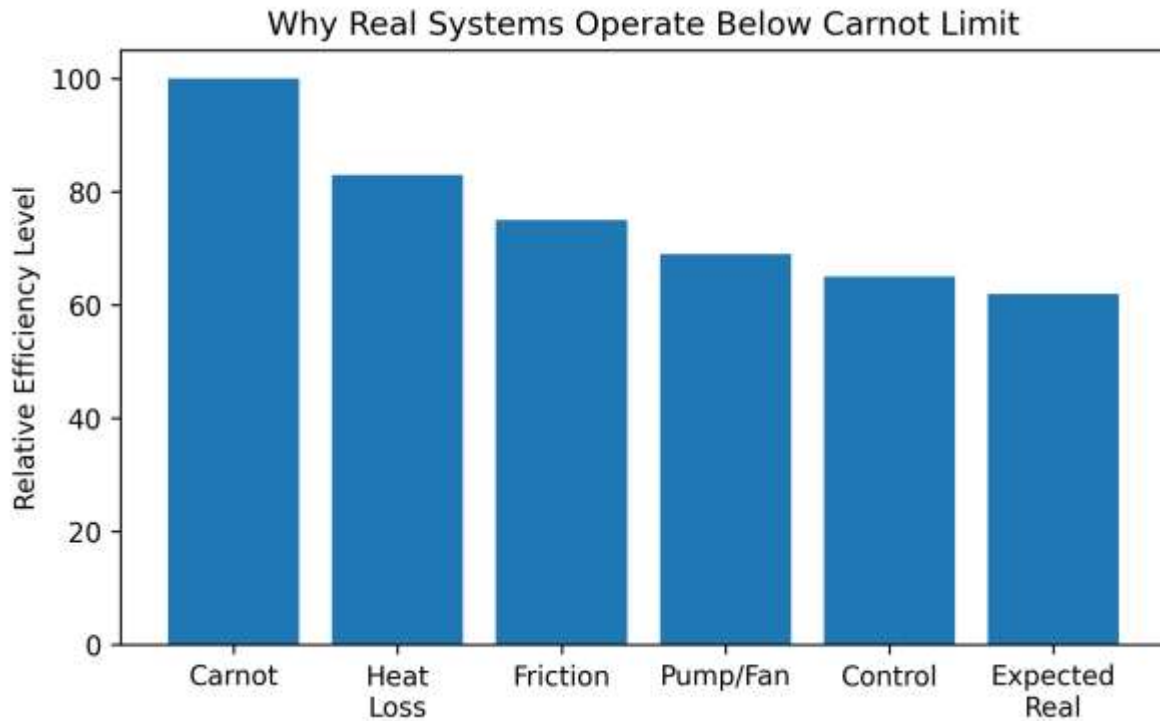


Figure 5. Why practical systems operate below Carnot efficiency.

6.1 Practical Meaning

The gap between actual efficiency and Carnot efficiency must be treated as a quantitative diagnostic space, not merely an abstract thermodynamic shortfall; every percentage point of gap represents recoverable energy that engineering intervention can partially reclaim.

- A large efficiency gap concentrated in the mechanical subsystem indicated by elevated bearing temperatures, abnormal vibration signatures or lubrication degradation identifies tribological losses as the primary corrective target, justifying investment in improved sealing, bearing redesign or surface treatment.
- A gap attributable to thermal losses, evidenced by high external surface temperatures or elevated heat exchanger outlet temperatures, directs the engineering team toward improved insulation specification, reduced wall conduction paths or enhanced heat exchanger surface area.
- Tracking the efficiency gap over time creates a degradation curve that supports predictive maintenance scheduling; a sustained increase in the gap between consecutive test campaigns, under identical boundary conditions, is a statistically reliable early indicator of fouling, seal wear or bearing deterioration before catastrophic failure occurs.
- The Second Law efficiency defined as the ratio of actual efficiency to Carnot efficiency provides a single dimensionless indicator of thermodynamic quality that is independent of system size, working fluid and operating pressure, enabling objective comparison across entirely different machine configurations within the same product portfolio.

- When the efficiency gap is decomposed into loss fractions and plotted as a Sankey diagram, it communicates the energy balance transparently to non-specialist stakeholders including procurement, quality assurance and regulatory reviewers, supporting informed decision-making without requiring thermodynamic expertise from every audience.
- Any design modification that claims to reduce the efficiency gap must be validated by repeating the full test protocol under identical boundary conditions; a reduction in the gap that cannot be reproduced across three independent test runs should be classified as measurement variability rather than genuine thermodynamic improvement, preventing false conclusions from being embedded in the design record.

7. Use of Carnot Limit for Mechanical and Purification Devices

7.1 How It Helps During Design

The Carnot limit establishes an inviolable thermodynamic ceiling during the conceptual design phase, allowing the design team to immediately reject efficiency targets that violate the second law before any detailed engineering or capital expenditure is committed.

- It enables early-stage temperature boundary optimisation by quantifying the theoretical efficiency gain achievable per kelvin of hot-side temperature increase, allowing thermal engineers to balance efficiency improvement against the corresponding increase in material stress, corrosion rate and pressure containment requirements.
- It provides a vendor-independent benchmark for evaluating supplier efficiency claims during procurement; any quoted efficiency figure approaching or exceeding the Carnot value calculated from the supplier's own stated operating temperatures must trigger a formal technical clarification request before the component is accepted.
- It guides the selection of working fluids and cycle configurations in thermal machines by allowing engineers to model the theoretical Carnot efficiency at the intended operating temperatures and compare it against published data for candidate fluids, identifying which thermodynamic cycles offer the best alignment between practical and ideal performance at those boundary conditions.
- It supports design-for-maintainability decisions by quantifying how much efficiency penalty is acceptable from fouling, seal degradation or bearing wear before the system drops below its contractual performance floor, directly informing service interval schedules and spare parts inventory planning.
- It is directly applicable to purification-related devices including membrane systems and adsorption columns by establishing the minimum theoretical energy cost of the separation process as a function of the temperature differential across the thermal regeneration stage, providing a lower-bound benchmark for energy consumption that separates thermodynamic limits from engineering implementation losses.
- It strengthens design review documentation by providing a clear, auditable thermodynamic justification for the chosen efficiency target, demonstrating to regulators, clients and quality

auditors that the performance specification is physically grounded and not based on empirical optimism or competitive pressure alone.

7.2 Practical Efficiency Target Setting

Step	Action	Reason
1	Measure hot and cold temperatures	Defines theoretical boundary
2	Calculate Carnot efficiency	Shows maximum possible efficiency
3	Measure actual efficiency	Shows real performance
4	Compare actual vs ideal	Identifies performance gap
5	Find dominant losses	Guides design improvement
6	Repeat after changes	Confirms improvement

7.3 Human-Readable Interpretation

A simple interpretation is that no real device can convert all heat into useful work because some energy must always move to the colder side.

8. Integration Methodology

8.1 Data Collection Plan

- Use calibrated temperature sensors with known measurement uncertainty, preferably RTD Class A or Type K thermocouple with cold junction compensation, positioned directly at the thermal boundary interfaces rather than in the bulk fluid stream.
- Measure real input power at the system boundary using a calibrated power analyser with true RMS capability, capturing harmonic content and power factor to avoid underestimating total electrical energy consumption in variable-frequency drive systems.
- Record useful output energy accurately using torque-speed transducers for shaft work or calibrated flow and differential pressure instruments for hydraulic work, applying appropriate time integration over the full test duration to capture transient behaviour.
- Log ambient conditions including dry-bulb temperature, relative humidity and barometric pressure at minimum five-minute intervals throughout testing, as these affect cold-side sink temperature and working fluid properties in open-circuit configurations.
- Repeat testing across a minimum of three distinct load points covering 50%, 75% and 100% of rated capacity to characterise part-load efficiency behaviour and identify the operating point of peak thermodynamic performance.
- Record maintenance condition before each test campaign, including time since last service, fouling index of heat exchangers, bearing condition, seal leakage status and lubricant grade, so that efficiency degradation trends can be correlated with specific maintenance variables over time.
- Conduct a measurement system analysis (MSA) prior to formal data collection, verifying that combined sensor, signal conditioning and data acquisition uncertainty is below 0.5% of full-

scale reading to ensure the efficiency gap analysis is not dominated by instrument error rather than genuine thermodynamic losses.

8.2 Calculation Pipeline

Calculation Item	Formula / Method
Kelvin Conversion	$T(K) = T(^{\circ}C) + 273.15$
Carnot Efficiency	$1 - T_c/T_h$
Actual Efficiency	Useful Output/Input Energy
Efficiency Gap	Carnot - Actual
Relative Performance	Actual/Carnot

8.3 Documentation Method

Every test should record operating condition, input energy, output energy, Carnot efficiency, actual efficiency and suspected losses.

9. Risks, Limitations and Mitigation Strategies

Risk	Possible Problem	Mitigation
Using Celsius	Incorrect efficiency result	Convert to Kelvin
Wrong System Boundary	Misleading efficiency comparison	Clearly define boundaries
Sensor Drift	Unreliable trend analysis	Calibration and reference checks
Ignoring Auxiliary Power	Efficiency appears unrealistically high	Include pumps/fans/controllers
Overheating	Material damage	Respect thermal safety margins
Fouling	Efficiency degradation over time	Preventive maintenance

9.1 Limitations of Carnot Analysis

The Carnot equation directly applies only to ideal reversible heat engines operating between two fixed-temperature reservoirs; real systems with variable hot-side temperatures, internal regeneration, or multi-stage compression and expansion require extended thermodynamic models such as the endoreversible engine framework or finite-time thermodynamics to produce accurate efficiency predictions.

- Pump efficiency is governed primarily by hydraulic, volumetric and mechanical loss mechanisms rather than by heat exchange between temperature reservoirs; applying the Carnot formula to a pump without establishing a clearly defined thermal boundary and temperature differential will produce a physically meaningless efficiency ceiling that cannot be used for engineering decisions.
- Membrane separation efficiency depends on partial pressure differentials, osmotic pressure gradients and molecular selectivity rather than on macroscopic temperature differences;

Carnot analysis must be supplemented with solution-diffusion or pore-flow models to provide a meaningful thermodynamic minimum energy benchmark for membrane systems.

- The Carnot model assumes perfectly reversible, quasi-static processes executed at infinitely slow speed; real machines must operate at finite speed to produce useful power output, meaning the Curzon-Ahlborn efficiency at maximum power approximately $1 - \sqrt{T_c/T_h}$ is a more realistic upper bound for high-throughput industrial applications than the classical Carnot efficiency.
- Carnot analysis does not account for the energy cost of auxiliary systems including control electronics, cooling circuits, lubrication pumps and instrumentation; when these parasitic loads constitute a significant fraction of total input energy as is common in small-scale or low-load systems the effective system efficiency may fall substantially below even conservative engineering targets without any failure in the primary thermodynamic cycle.
- The two-reservoir model assumes perfectly isothermal heat addition and rejection; in practice, heat exchangers operate with finite temperature differences and logarithmic mean temperature driving forces, introducing irreversibilities that lower the effective temperature ratio seen by the working fluid and reduce achievable efficiency below the nominal Carnot value calculated from bulk reservoir temperatures.
- Carnot analysis provides no information about transient performance, start-up losses, thermal mass effects or cyclic degradation; a system that achieves satisfactory Second Law efficiency under steady-state conditions may still exhibit poor energy performance during start-up, load-following or shutdown phases, requiring separate transient energy analysis to characterise whole-life thermodynamic behaviour accurately.

9.2 Recommended Mitigation Strategy

Carnot analysis must be integrated with fluid mechanics modelling to correctly account for pressure-drop losses, flow regime transitions and compressibility effects that directly reduce shaft work output independently of the thermal cycle efficiency, ensuring that the total efficiency model reflects the full loss budget of the system.

- Material science inputs must be incorporated when evaluating the upper operating temperature boundary; the Carnot efficiency gain from increasing T_h must be weighed against accelerated creep, oxidation and fatigue in structural components, with the thermodynamically optimal temperature selected only after confirming that material lifetime at that temperature meets the required service interval.
- Control-system analysis must be layered onto the thermodynamic model to evaluate how control loop bandwidth, sensor latency and actuator response time affect the ability to maintain stable temperature boundary conditions; a thermodynamically optimal design that is unstable under closed-loop control will not deliver predicted efficiency in field operation.
- Where the system boundary includes non-heat-engine subsystems such as pumps, membranes or adsorption beds, subsystem-specific efficiency models pump affinity laws, solution-diffusion permeability equations or adsorption isotherm models must be coupled

with the Carnot framework to produce a complete and physically consistent whole-system efficiency estimate.

- Sensitivity analysis should be performed on all input parameters to the Carnot model particularly T_h and T_c to quantify how measurement uncertainty propagates into efficiency uncertainty; if the combined sensor uncertainty produces an efficiency confidence interval wider than the claimed performance margin, the measurement system must be upgraded before conclusions can be drawn with engineering rigour.
- Finite-time thermodynamic models, particularly the Curzon-Ahlborn endoreversible engine model, should be applied alongside classical Carnot analysis whenever the system operates at high power density or fast cycle rates, as these conditions push the achievable efficiency significantly below the Carnot ceiling and the more realistic finite-time bound provides a better engineering target for optimisation studies.
- All assumptions underpinning the Carnot analysis system boundary definition, steady-state operating conditions, reservoir temperature values, excluded parasitic loads must be explicitly documented in the engineering file and subjected to formal assumption review at each design gate, ensuring that changes in system scope or operating conditions trigger a re-evaluation of the thermodynamic benchmark rather than allowing an outdated Carnot reference to persist through subsequent design stages.

10. Validation and Testing Strategy

10.1 Test Objectives

- Confirm that actual efficiency never exceeds the Carnot efficiency calculated from simultaneously measured hot-side and cold-side temperatures, and treat any violation as an immediate trigger for instrument calibration audit and system boundary re-examination before further data are recorded.
- Establish measurement repeatability by conducting a minimum of three identical test runs under controlled steady-state conditions and calculating the coefficient of variation for each efficiency metric; a coefficient of variation exceeding 2% indicates unacceptable instrument noise or process instability that must be resolved before formal performance data are used in design decisions.
- Characterise efficiency response across a defined operating envelope spanning at least three hot-side temperatures and three load levels, generating a two-dimensional efficiency map that reveals the thermodynamically optimal operating region and informs control strategy development for field deployment.
- Identify and quantify dominant losses by allocating the measured efficiency gap fractionally across all identified loss mechanisms using a structured energy balance, producing a ranked loss attribution table that directly prioritises engineering improvement actions by their potential efficiency recovery value.
- Compare each design iteration against the established baseline using the Second Law efficiency as the primary normalised metric, ensuring that component-level changes seal

upgrades, heat exchanger fouling removal, insulation improvements produce statistically significant and reproducible improvements in the efficiency ratio before being accepted into the production configuration.

- Conduct a long-duration endurance test spanning a representative service interval to measure the rate of efficiency degradation as a function of operating hours, fouling accumulation and thermal cycling, providing quantitative data to support the definition of preventive maintenance intervals and contractual performance guarantees.
- Validate the complete measurement chain from sensor to final efficiency calculation by introducing known reference conditions traceable temperature sources, precision power loads and certified flow standards at the start of every test campaign, confirming that the calculated Carnot and actual efficiency values are free from systematic instrument bias before they are entered into the engineering record.

10.2 Suggested Test Matrix

Test	Variable Changed	Data to Record	Expected Learning
Temperature Sweep	Hot-side temperature	Th,Tc,input,output	Effect of temperature difference
Load Test	Mechanical load	Power,flow,pressure	Performance under demand
Insulation Test	Insulation quality	Heat loss,surface temperature	Impact of leakage
Long-run Test	Operating time	Efficiency trend,fouling	Degradation over time
Maintenance Test	Before/after cleaning	Pressure drop,energy use	Effect of maintenance

10.3 Acceptance Criteria

Measurement repeatability must be demonstrated by a coefficient of variation below 2% across three independent test runs at the same operating point, with outliers investigated and attributed to a specific assignable cause before the test series is accepted as valid.

- Calculated actual efficiency must remain strictly below the Carnot efficiency value computed from simultaneously measured boundary temperatures throughout every test run; a single exceedance, even within measurement uncertainty bounds, must be treated as a non-conformance requiring root cause analysis before the test result is used in any engineering report.
- All efficiency calculations must be fully traceable, with raw sensor readings, conversion factors, integration periods, parasitic load exclusions and formula applications documented in a calculation record that can be independently audited and reproduced by a third party without additional information from the test team.
- The combined measurement uncertainty of the efficiency result, calculated by propagating individual sensor uncertainties through the efficiency formula using standard uncertainty

analysis per ISO GUM methodology, must be documented and must not exceed half of the claimed performance margin above the minimum contractual efficiency specification.

- The loss attribution breakdown must account for a minimum of 90% of the total measured efficiency gap, with each attributed loss mechanism supported by at least one independent measurement or physical model; an unattributed residual exceeding 10% of the gap must be flagged as an open technical finding and investigated before the design is closed out.
- Efficiency values recorded at part-load operating points 50% and 75% of rated capacity must fall within the predicted range defined by the efficiency model developed during the design phase; deviations exceeding the model prediction by more than 5 percentage points at any tested load point must trigger a model revision and a formal technical review before part-load operation is endorsed.
- All test documentation including raw data files, calibration certificates, ambient condition logs, equipment identification records and the signed test protocol must be archived in the project technical file within five working days of test completion, ensuring the complete evidence base is available for design review, customer witness inspection and regulatory submission without reliance on the memory of individual test personnel.

11. Documentation and Compliance Checklist

Checklist Item	Requirement
Temperature Units	Use Kelvin
Formula	$\eta = 1 - T_c/T_h$
Boundary Definition	Clearly define energy inputs and outputs
Loss Analysis	Include thermal, mechanical and electrical losses
Figures	All diagrams require captions
References	Provide direct URLs
Claims	Do not exceed Carnot limit

11.1 Terms Used in This Report

Term	Simple Meaning
Heat Engine	System converting heat into work
Hot Reservoir	High-temperature heat source
Cold Reservoir	Low-temperature sink
Carnot Efficiency	Maximum theoretical efficiency
Irreversibility	Real-world process causing losses
Efficiency Gap	Difference between actual and ideal

12. Conclusions and Next Steps

The Carnot limit is the single most important thermodynamic reference for heat-based systems; it establishes an inviolable upper efficiency boundary derived directly from the second law of thermodynamics, making it the authoritative benchmark against which every efficiency claim, design target and test result in this project must be evaluated.

- Real systems invariably operate below the theoretical Carnot limit because every practical machine contains irreversibilities friction, turbulence, heat leakage, electrical conversion loss and fouling whose combined effect grows with operating intensity and system age; the engineering objective is not to eliminate this gap, which is physically impossible, but to understand, quantify and systematically reduce it to the extent that is economically and technically justified.
- The Second Law efficiency the ratio of actual to Carnot efficiency has been established in this report as the primary normalised performance indicator for the system, enabling rigorous cross-comparison between design iterations, operating conditions and competing configurations on a single, physically meaningful scale that is independent of system size and working fluid.
- The efficiency gap analysis methodology developed in this report encompassing sensor specification, measurement system analysis, loss attribution mapping and longitudinal trend monitoring provides the project team with a complete, repeatable and auditable process for converting raw test data into actionable engineering decisions throughout the system's development and operational life.
- The limitations of Carnot analysis identified in Section 9 its inapplicability to non-heat-engine subsystems, its assumption of quasi-static reversible processes, and its insensitivity to transient and part-load behaviour must be actively managed by integrating complementary engineering models including finite-time thermodynamics, subsystem-specific efficiency correlations and control-system dynamic analysis into the overall performance prediction framework.
- The risk mitigation strategies documented in Section 9 correct temperature unit conversion, precise system boundary definition, inclusion of auxiliary loads, calibrated instrumentation and preventive maintenance scheduling must be implemented as mandatory engineering controls rather than advisory recommendations, as failure to apply any one of them can invalidate the entire efficiency analysis and lead to incorrect design decisions.
- The immediate next step is to apply the complete Carnot analysis framework defined in this report to real prototype test data, generate the first quantitative loss attribution map, identify the highest-priority loss mechanisms for the current design configuration, and present a costed improvement proposal to the project review board with projected efficiency gains, implementation timelines and verification test plans for each proposed modification.

12.1 Recommended Next Actions

1. Collect actual hot-side and cold-side operating temperatures at the true thermal boundary interfaces using calibrated instruments, and document the measurement locations with reference to system P&ID drawings to ensure reproducibility across test campaigns.
2. Convert all temperature values to Kelvin before applying the Carnot equation, and include a cross-check by computing efficiency at both boundary conditions to detect sensor placement errors or systematic offsets introduced by fluid stratification near the measurement point.

3. Measure input and useful output energy over a stable steady-state operating window of sufficient duration to reduce stochastic measurement noise below 1% of the mean value, using integrated energy rather than instantaneous power readings.
4. Calculate actual efficiency and compare against the Carnot efficiency to derive the absolute efficiency gap, then compute the Second Law efficiency (actual / Carnot) as the primary normalised performance indicator for cross-system benchmarking.
5. Identify dominant losses by constructing a quantitative loss attribution map, allocating the efficiency gap fractionally across friction, heat leakage, pressure drop, electrical conversion loss and fouling, and rank each by its contribution to prioritise the engineering improvement programme.
6. Implement targeted design or maintenance modifications addressing the highest-ranked loss mechanism first, document the change formally in the engineering change log, and define acceptance criteria specifying the minimum measurable efficiency improvement required to validate each modification.
7. Repeat the full test protocol after each modification under identical boundary conditions, update the loss attribution map with new measurements, and maintain a longitudinal efficiency trend record to support reliability analysis and predictive maintenance scheduling over the system's operational life.

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Appendix A - Assumptions

- The system contains at least one thermal process in which heat is transferred between a high-temperature source and a low-temperature sink, producing or consuming useful mechanical, electrical or pressure-enthalpy work at a defined system boundary.
- Temperatures at both the hot-side and cold-side boundaries are measurable with calibrated instruments, convertible to absolute Kelvin values, and representative of the true thermodynamic boundary conditions experienced by the working fluid at the point of heat exchange rather than at remote measurement locations.
- The Carnot limit is treated exclusively as a theoretical upper efficiency boundary and not as an achievable engineering target; all practical efficiency specifications in this report are set

below the Carnot value by a margin that reflects known and estimated irreversibility contributions from all identified loss mechanisms.

- Real-world losses are present in all subsystems and no component is assumed to operate as an ideal, lossless element; every friction surface, heat transfer interface, fluid passage and electrical conversion stage is assumed to contribute a positive, non-zero loss that must be measured, estimated or bounded within the efficiency analysis.
- The system operates under sufficiently stable steady-state conditions during each test window that transient energy storage effects in thermal mass, fluid inventory and rotating inertia are negligible relative to the integrated input and output energy over the measurement period, so that time-averaged efficiency values are representative of the intended operating state.
- The system boundary is clearly and unambiguously defined before any efficiency calculation is performed, with all energy streams crossing the boundary including auxiliary electrical supplies, heat losses through casing walls, and mechanical work transmitted through seals or couplings — either included in the energy balance or explicitly excluded with documented technical justification.
- The measurement uncertainty associated with each instrument used in the efficiency calculation has been characterised, is traceable to a recognised national or international metrology standard, and has been combined using standard uncertainty propagation methods to produce a stated efficiency uncertainty that is reported alongside every efficiency result presented in this document.

Appendix B - Quick Calculation Template

Input / Output	Unit	Notes
Hot-side Temperature T_h	K	Convert from Celsius if needed
Cold-side Temperature T_c	K	Environmental sink
Carnot Efficiency	%	$(1 - T_c/T_h) \times 100$
Input Energy	J or kWh	Include all relevant sources
Useful Output Energy	J or kWh	Define output clearly
Actual Efficiency	%	Output/Input x100
Efficiency Gap	%	Carnot - Actual

Appendix C - Glossary

Word	Meaning
Absolute Temperature	Temperature measured in Kelvin
Reservoir	Body supplying or absorbing heat
Reversible Process	Ideal process without losses
Irreversible Process	Real process with losses
Thermal Efficiency	Useful work divided by heat input
Benchmark	Reference value for comparison

